

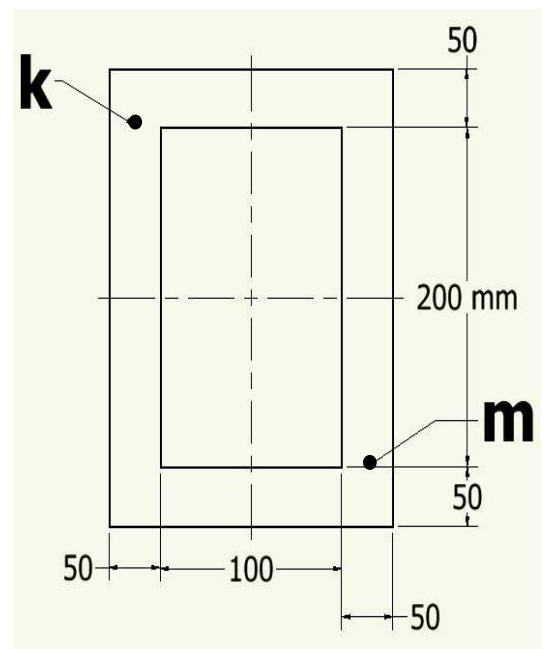
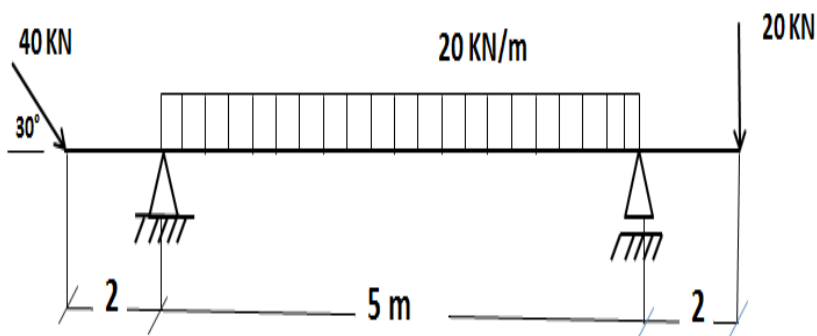
Student Name:

ID:

Sec:

Question (1) 10 Marks

1. Draw (N.F.D), (S.F.D) and (B.M.D) for the given structure. (4 Marks)
2. Calculate and draw the normal and shear stress diagrams at the critical section. (4 Marks)
3. Determine the state of stress at **m** & **k** showing the results on infinitesimal elements. (2 Marks)



Dims. in mms

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Question (2) 10 Marks

The plane steel structure shown in the figure is subjected to the given loads and has the shown cross section.

1. Calculate and draw the normal and shear stress diagrams at section **s-s**.

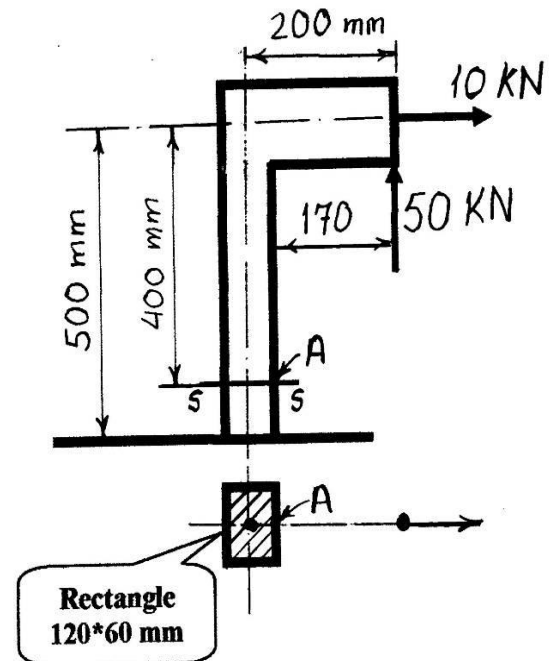
(6 Marks)

2. Determine the state of stress at point **A** showing the results on an infinitesimal element.

(2 Mark)

3. Find the values of the principal normal and maximum shear stresses at point **A**.

(2 Mark)



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Learning Outcomes (LOs)

Questions	LO1	LO2	LO3	LO4
Question 1	•	•		•
Question 2	•	•		•

Cognitive Domain	
LO1	Apply the basics of stress analysis in mechanical design.
LO2	Analyze the various loads and stresses acting on mechanical structures.
Psychomotor Domaine	
LO3	Implement actual methods for calculating deformations, slopes and deflections of mechanical structures.
Affective Domaine	
LO4	Combine stresses and strains for the design of mechanical components.